

Data Snooping and Market-Timing Rule Performance

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Abstract

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JEL classification: G11, G14.

Keywords: Market Timing, Data Snooping, Reality Check, SPA Test.

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Abstract

We reassess the profitability of market-timing rules when controlling for data-snooping biases. For the first time, a comprehensive set of simple and complex market-timing rules is examined and tested for statistical significance, using the White (2000) “Reality Check” and the Hansen (2005) SPA test. For the full sample period, we find that individual market-timing rules show significant outperformance when considered in isolation but that these results do not remain significant after correcting for data snooping. However, we find that during the subperiod from 1981 to 1994, certain complex rules indeed outperform a buy-and-hold strategy even after data-snooping correction is made.

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It has always been an investor's dream to time the market in order to earn excess returns. The methodology of market timing seems to be quite simple: Remain invested in the stock market when expected returns are high and switch to cash investments when the market is expected to underperform. The timing of the switch is indicated by signals based on fundamental or sentiment indicators. The potential of market-timing strategies to earn extraordinary profits is documented, for instance, by Shilling (1992), who found that, during the period from 1946 to 1991, investors could have increased their annual return from 11.2% to 19.0% by avoiding exposure to the stock market during the 50 weakest months and being long stocks the rest of the time. In contrast, missing the 50 strongest months in the stock market would have reduced annual returns to only 4% per annum. The problem, of course, is that detecting weak stock markets in time but not getting out too soon remains a difficult task.

Market-timing rules usually build on empirical findings of the ability of certain indicators to predict future stock market returns. The literature has identified many useful indicators, such as the earnings-to-price ratio (Campbell and Shiller (1988b), Campbell and Shiller (1998)), the dividend yield (Shiller (1984), Fama and French (1988)), the dividend-payout ratio (Lamont (1998)), the maturity spread and the credit spread (Campbell (1987), Fama and French (1989)), and the gilt-equity yield ratio (Clare et al. (1994), Brooks and Persaud (2001)).¹ However, if capital markets are efficient, market-timing rules should not be able to beat a benchmark, such as a buy-and-hold strategy.² In other words, publicly available information should not be useful for predicting future stock market movements.

When assessing the profitability of market-timing rules, it is critical to account for possible biases in statistical inference due to data snooping.³ The danger of data-snooping biases is especially acute whenever economic theory is vague about the functional form of the relationship between economic variables. In such a situation, researchers can apply a variety of market-timing rules to the same data set and uncover a subset of rules that turn out to

be profitable even though they would not generate superior performance out-of-sample. Lo and MacKinlay (1990) argue that data snooping might be the result of survivorship bias. That is, many rules that have been investigated in the past but have not generated profits superior to a benchmark are not published and are thus filtered out over time. Hence, the superior power of profitable rules is not due to any merit of the design of the rules. Because statistical inference that considers only the surviving rules will be biased, it is important to account for the full set of initial rules. The so-called “Reality Check” (RC), developed by White (2000), allows us to account for data-snooping biases when testing for possible superior performance of certain rules. The basic idea of this test is to draw statistical inference from an empirical distribution of a performance measure, considering the full universe of models (in our case, market-timing rules) from which the best rule is drawn. Hansen (2005) proposed the so-called test for superior predictive ability (SPA), which also allows for data-snooping correction and, under certain conditions, increases the test power of White’s “Reality Check.”

There is by now a vast literature reporting a variety of market-timing rules. For instance, Fisher and Statman (2006a) and Fisher and Statman (2006b) investigate the profitability of market timing based on financial ratios. Shen (2003) finds that the spread between the earnings-to-price ratio and a variety of prevailing interest rates can be used to successfully time the market. Studies of the ability of various interest rates to time the market include Breen et al. (1989), Prather and Bertin (1997), and Prather and Bertin (1998). The performance of market-timing rules based on index volatility changes is investigated, for instance, by Copeland and Copeland (1999). However, the studies on market-timing performance lack appropriate corrections for data-snooping biases. The purpose of our paper is therefore to assess the profitability of a large number of market-timing rules and to examine possible data-snooping biases. Our study is in the spirit of Sullivan et al. (1999)’s and Hsu and Kuan (2005)’s evaluations of the profitability of technical trading. They construct their rules based on historical prices, but do not include information from indicators based on fundamentals and investor sentiment. Both studies show that, in certain markets and during

certain periods, technical trading is useful to beat a benchmark. However, Sullivan et al. (1999) show that out-of-sample the superior performance of the best rules is no longer significant. Hsu and Kuan (2005) find that the best technical-trading rules are only capable of beating the benchmark in “young markets” such as the NASDAQ Composite and the Russell 2000. Sullivan et al. (2001), in a similar experiment, argue that calendar rules do not show superior performance when adjustment for data-snooping biases is made. In yet another study on the profitability of technical-trading rules, Qi and Wu (2006) apply the RC methodology to examine the profitability of such trading rules in the foreign-exchange market and find that the significant outperformance of the best rules (when considered in isolation) is not robust to data-snooping effects.

Our paper is the first to quantify the possible data-snooping biases in the market-timing literature and to test whether the considered market-timing rules are truly superior to a buy-and-hold strategy. For this purpose, we use White (2000)’s RC methodology and Hansen (2005)’s SPA. As opposed to the previously mentioned studies on data-snooping biases in the context of technical trading, the rules in this paper are not constructed based on historical prices, but are based on a set of fundamental and sentiment indicators. We construct two sets of timing rules: a basic set of simple rules, each based on only one indicator, and an extended set which also includes complex rules based on information from various indicators.⁴ Each universe of market-timing rules includes both rules described in the market-timing literature and rules that we construct ourselves based on indicators that were previously found to predict stock market returns.⁵ For each rule, we employ various parameterizations, which gives us a comprehensive universe of 6,792 simple rules. The extended set comprises 8,768 rules.⁶ As performance measures, we use the average return and the Sharpe ratio.

The results of our experiments suggest that, when considered in isolation, the performance of certain market-timing rules from both sets may indeed be highly statistically significant. However, once the effects of data snooping are appropriately accounted for, the best market-

timing rules from the basic set are no longer significant for any period. For the full sample period and for the subperiod from 1995 to 2007, we find the same insignificant result for the extended set of rules. Only for the subperiod from 1981 to 1994 does the outperformance of the best rule from the extended set remain significant. The results are quite similar for both performance measures. Furthermore, our findings are robust to the impact of transaction costs and to allowing for short selling the index.

The paper proceeds as follows: Section 1 explains the testing procedures (RC and SPA) used to correct for data-snooping biases. Section 2 discusses the intuition and specification of the market-timing rules to be tested. Section 3 describes the data. Section 4 shows the empirical results for the basic set of rules. Section 5 shows the empirical results for the extended set of rules. Section 6 presents some robustness checks on the results. Section 7 concludes.

1. Testing Procedures: The “Reality Check” and the SPA Test

The dangers of data snooping have long been recognized as a serious problem of empirical studies in finance (see, for instance, Lo and MacKinlay (1990), Brock et al. (1992), and Ferson et al. (2003)). As we are investigating a large universe of market-timing rules, a robust methodology to avoid spurious statistical inference due to data snooping is needed. We therefore employ the “Reality Check” (RC), introduced by White (2000), and the test for superior predictive ability (SPA), introduced by Hansen (2005). Both procedures allow for intensive search for models while ensuring that the results are robust and do not result from mere chance. Both procedures build on the work of Diebold and Mariano (1995) and West (1996). In this section, we briefly outline both testing procedures and refer the reader to the original articles for rigorous derivations.

The RC tests the null hypothesis that the best model does not have superior predictive ability over a benchmark model, while taking into account the full set of models, against the alternative that the best model does have superior predictive ability. The test is based on the $l \times 1$ performance statistic,

$$\bar{\mathbf{f}} = \frac{1}{n} \sum_{t=R}^T \hat{\mathbf{f}}_{t+1}, \quad (1)$$

where l is the number of market-timing rules and n is the number of prediction periods indexed from R through T , so that $T = R + n - 1$. $\hat{\mathbf{f}}_{t+1} = f(\mathbf{Z}_t, \hat{\boldsymbol{\beta}}_t)$ is the performance measure, where $\mathbf{Z}_t$ is a matrix which contains a vector of dependent variables and a vector of predictor variables and $\hat{\boldsymbol{\beta}}_t$ is a vector of estimated parameters. It is assumed that these parameters satisfy the conditions of Diebold and Mariano (1995) and West (1996), so that parameter uncertainty vanishes asymptotically. In our experiment, we consider various parameterizations of each trading rule (β_k , $k = 1, \dots, l$). The parameterizations directly produce returns, so there are no estimated parameters.

We use the returns generated from the l timing rules as a performance measure.⁷ For a timing rule k , we follow Sullivan et al. (1999) and specify $f_{k,t+1}$ as

$$f_{k,t+1} = \ln \left[1 + \frac{S_{t+1} - S_t}{S_t} X_k(\mathbf{Z}_t, \boldsymbol{\beta}_k) \right] - \ln \left[1 + \frac{S_{t+1} - S_t}{S_t} X_0(\mathbf{Z}_t, \boldsymbol{\beta}_0) \right], \quad k = 1, \dots, l, \quad (2)$$

where $\mathbf{Z}_t$ consists of the predictor variables (described in Section 2) and $\boldsymbol{\beta}_k$ denotes the different parameterizations of the timing rules; subscript 0 refers to the benchmark model. S_t is the price of the S&P index at time t . X_k and X_0 are “timing functions” which take on the value 1 for “invest in the stock market” and 0 for “hold cash.” Based on this performance statistic, we test whether there is a timing rule that delivers superior performance over a simple buy-and-hold strategy, where $X_0 = 1$ at all times. Formally, the null hypothesis is:

$$H_0 : \max_{k=1,\dots,l} \{E(f_k)\} \leq 0. \quad (3)$$

If the null can be rejected, it has been established that a timing rule exists that outperforms the benchmark. It has been shown by White (2000) that, under weak assumptions about the stationarity, dependence structure, and moments of $\hat{\mathbf{f}}_t$, the distribution of the test statistic can be obtained by applying the stationary bootstrap of Politis and Romano (1994) as follows. In step 1, for each timing rule $k = 1, \dots, l$, we generate a resample of $\{f_{k,t+1}\}_{t=R}^{t=T}$ by drawing (geometrically distributed) blocks from the observed return series, with mean block length $1/q$.⁸ We shall denote the resampled series by $f_{k,t+1,j}^*$, where subscript j indicates the j -th repetition of the bootstrap. We repeat the process J times. In step 2, we calculate the mean of the bootstrapped return series, $\bar{f}_{k,j}^* = n^{-1} \sum_{t=R}^T f_{k,t+1,j}^*$, $\forall k = 1, \dots, l$. In step 3, we compute the following statistics:

$$V_{RC} = \max_{k=1,\dots,l} \{\sqrt{n} \bar{f}_k\}, \quad (4)$$

$$V_{RC,j}^* = \max_{k=1,\dots,l} \{\sqrt{n}(\bar{f}_{k,j}^* - \bar{f}_k)\}, \quad j = 1, \dots, J. \quad (5)$$

We then compare V_{RC} with the quantiles of $V_{RC,j}^*$. A p-value of the RC is computed as

$$p_{RC} = \sum_{j=1}^J \frac{\mathbb{1}_{\{V_{RC,j}^* > V_{RC}\}}}{J}, \quad (6)$$

where $\mathbb{1}_{\{\cdot\}}$ denotes the indicator function. In our empirical analysis, we set $J = 1000$ and choose a smoothing parameter of $q = 0.5$.⁹

Hansen (2005)'s SPA is very similar to the RC, yet it includes some refinements that can improve the test power in most cases. The SPA makes use of the following studentized test statistic:

$$V_{SPA} = \max \left[\max_{k=1, \dots, l} \frac{\sqrt{n} \bar{f}_k}{\hat{\sigma}_k}, 0 \right], \quad (7)$$

where $\hat{\sigma}_k^2$ is a consistent estimate of $\sigma_k^2 = \text{var}(\sqrt{n} \bar{f}_k)$. We employ the estimator given in Hansen (2005), who also suggests invoking a different null distribution based on $N(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Omega}})$, where $\hat{\boldsymbol{\Omega}}$ denotes a consistent estimate of the asymptotic covariance matrix of $\bar{\mathbf{f}}$ and $\hat{\boldsymbol{\mu}}$ is an estimate for $E(\mathbf{f}_t)$. Hansen (2005) advocates the use of the following estimator:

$$\mu_k = \bar{f}_k \mathbb{1}_{\{\sqrt{n}(\bar{f}_k/\hat{\sigma}_k) \leq -\sqrt{2 \ln \ln n}\}}. \quad (8)$$

By choosing this estimator, we make sure that irrelevant models do not asymptotically influence the distribution of the test statistic. This can be shown by applying the law of the iterated logarithm, which ensures that $\frac{\sqrt{n} \bar{f}_k - \mu_k}{\sigma_k}$ stays within certain bounds with probability 1 asymptotically.

The implementation of the SPA is also very similar to that of the RC. In step 1, for each timing rule $k = 1, \dots, l$, we generate a resample of $\{f_{k,t+1}\}_{t=R}^{t=T}$ by drawing (geometrically distributed) blocks from the observed return series. We shall denote the resampled series $\{f_{k,t+1,j}^*\}$, where subscript j indicates the j -th repetition of the bootstrap. In step 2, we calculate $Z_{k,t+1,j}^* = f_{k,t+1,j}^* - \bar{f}_k \mathbb{1}_{\{\sqrt{n}(\bar{f}_k/\hat{\sigma}_k) \geq -\sqrt{2 \ln \ln n}\}}, \forall k = 1, \dots, l, t = R, \dots, T$. In step 3, we compute the following statistics:

$$V_{SPA} = \max \left[\max_{k=1, \dots, l} \frac{\sqrt{n} \bar{f}_k}{\hat{\sigma}_k}, 0 \right], \quad (9)$$

$$V_{SPA,j}^* = \max \left[\max_{k=1, \dots, l} \frac{\sqrt{n} \bar{Z}_{k,j}^*}{\hat{\sigma}_k}, 0 \right], \quad j = 1, \dots, J, \quad (10)$$

where $\bar{Z}_{k,j}^* = n^{-1} \sum_{t=R}^T Z_{k,t+1,j}^*$. We then compare V_{SPA} with the quantiles of $V_{SPA,j}^*$. A p-value of the SPA is given by

$$p_{SPA} = \sum_{j=1}^J \frac{\mathbb{1}_{\{V_{SPA,j}^* > V_{SPA}\}}}{J}. \quad (11)$$

Throughout the paper, we refer to this p-value as the consistent SPA p-value (SPA-c). In addition, we calculate an inconsistent lower bound for the consistent SPA p-value, called SPA-l, which Hansen (2005) computes by replacing $\bar{Z}_{k,j}^*$ in equation (10) with $\bar{Z}_{k,j}^{l*}$, the average of the bootstrapped series $Z_{k,t+1,j}^{l*} = f_{k,t+1,j}^* - \max[\bar{f}_k, 0]$.

For both testing procedures, we use the Sharpe ratio, in addition to raw returns, as a performance measure. If the Sharpe ratio is used as a performance measure, the null hypothesis is:

$$H_0 : \max_{k=1,\dots,l} \{g(E(h_k))\} \leq g(E(h_0)), \quad (12)$$

where $\mathbf{h}$ is a 3×1 vector given by

$$h_{k,t+1}^1 = \frac{S_{t+1} - S_t}{S_t} X_k(\mathbf{Z}_t, \boldsymbol{\beta}_k), \quad (13)$$

$$h_{k,t+1}^2 = \left(\frac{S_{t+1} - S_t}{S_t} X_k(\mathbf{Z}_t, \boldsymbol{\beta}_k) \right)^2, \quad (14)$$

$$h_{k,t+1}^3 = r_{t+1}^f; \quad (15)$$

r_{t+1}^f is the risk-free interest rate at time $t + 1$ and $g(\cdot)$ is given by

$$g(E(h_{k,t+1})) = \frac{E(h_{k,t+1}^1) - E(r_{t+1}^f)}{\sqrt{E(h_{k,t+1}^2) - (E(h_{k,t+1}^1))^2}}. \quad (16)$$

The expectations are estimated by the sample mean. We then construct the relevant statistic

$$\bar{f}_k = g(\bar{h}_k) - g(\bar{h}_0), \quad (17)$$

where $\bar{h}_k$ and $\bar{h}_0$ are arithmetic averages calculated over the sample period for trading rule k and the buy-and-hold strategy. The application of the bootstrap procedure to the difference of Sharpe ratios works similarly to the procedure described for the difference in raw returns (for details, see Sullivan et al. (1999), p. 1653).

2. Specification of Market-Timing Rules

In our investigation, we replicate the attempts of academics and the investment industry to find profitable market-timing rules. For a meaningful assessment of data-snooping biases, we have to construct a sufficiently large set of market-timing rules from which rules that have been applied or considered could possibly have been drawn. Sullivan et al. (1999) and Sullivan et al. (2001) point out that the selection of rules to be included in the analysis is especially important, since data-snooping biases are only corrected for relative to the universe of rules investigated. In particular, “the magnitude of data-snooping effects on the assessment of the performance of the best trading rule is determined by the dependence between all the trading rules’ payoffs, so the design of the universe from which the trading rules are drawn is crucial to the experiment” (Sullivan et al. (1999), p. 1654). When conducting the data-snooping tests, two possible issues can arise. On the one hand, considering too many “irrelevant” rules dilutes the power of the test and could render genuinely significant rules insignificant (see Hansen (2005)).¹⁰ This is why we discard trading rules based on new technology which has not been available for most of the sample period. On the other hand, if too few market-timing rules are included, the data-snooping correction would not account for the full set of rules from which effective rules conceivably could have been drawn; our estimated p-values would therefore be biased towards zero. Lo and MacKinlay (1990) refer to this as the “file-drawer” problem: Many rules that do not generate superior returns do not get published and are therefore no longer considered. It is therefore important not only to include the rules/parameterizations that have been publicly reported, but also other rules/parameterizations that may have been considered, since the successful rules are drawn from the entire set of successful and unsuc-

cessful rules. For those reasons, we find a balance and choose a fairly large variety of market-timing rules and parameterizations that investors and academics may have considered. In a first experiment, we consider a set of 6,792 simple market-timing rules, each based on one indicator only (basic set). Our second universe of rules adds complex strategies to the basic set and spans a universe of 8,768 market-timing rules (extended set). Even though our sets of rules do not exhaust all the rules/parameterizations that have ever been applied, our experiment covers a reasonably large universe of the most common and most important trading rules and is by far the most comprehensive in the field of market timing. In our subsample from 1995 to 2007, the number of rules increases to 8,280 for the basic set and 10,256 for the extended set, since during this time period we include additional indicators based on the availability of data. (See Section 3 for details on the construction of indicators.)

In the following section, we describe the rules applied in our analysis. In order to trigger switching signals, several threshold values of the indicators are considered. Thresholds are either a historical value, such as the moving average or percentile (those are referred to as standard thresholds) or a fixed number (see Section 3 for details). The general assumption for all market-timing rules is that an investor is usually long the index but tries to temporarily exit the market and switch to holding cash in bad times, those being indicated by the timing indicators.

2.1. Rules Based on Financial Ratios

The indicators for market timing that have received the most attention in the literature are the financial ratios: the dividend yield, the earnings-to-price (E/P) ratio, and the book-to-market (B/M) ratio. Many investors assume that stock prices do not wander too far from their normal levels relative to such fundamental values as earnings, dividends, and book value, but revert back to their normal levels after periods of deviation. That is, financial ratios are assumed to fluctuate within historical ranges and to be positively related to expected returns. Return predictability by means of the book-to-market ratio is investigated, for instance, by

Kothari and Shanken (1997) and Pontiff and Schall (1998). Shiller (1984), Campbell and Shiller (1988a), Campbell and Shiller (1988b), Fama and French (1988), Fama and French (1989), and Campbell and Shiller (2001), among others, argue that the dividend yield and the E/P ratio have predictive power. Based on these findings, Brennan et al. (1997) and Campbell and Viceira (1999) discuss the extent to which the predictive powers of the dividend yield and the E/P ratio are useful for timing the market. Fisher and Statman (2006a) and Fisher and Statman (2006b) also investigate the success of market timing based on financial ratios and find that the performance of buy-and-hold investors is usually higher than that of various timing rules based on E/P ratios and dividend yields.

2.1.1. Earnings-to-Price Ratio (E/P Ratio)

As outlined above, it is reasonable to assume that the E/P ratio generally follows a tendency of mean reversion; stock prices and the corresponding indicator of the fundamental value (earnings) are not expected to wander too far off. Considering the E/P ratio as the relative price of a stock, the economic rationale for using this ratio to time the market is quite simple. Stocks with a low E/P ratio provide low earnings relative to their price and are thus considered to be expensive. If the aggregate E/P ratio falls below a certain threshold, the market is considered to be overvalued and investors therefore switch from holding the index to cash. The rule applied in our investigation is specified as follows: When the E/P ratio falls below a certain threshold x , investors switch from being invested in the index to holding cash. When the ratio rises above x , investors revert to holding the index.

2.1.2. Dividend Yield

Market timing based on the dividend yield is quite similar to market timing by means of the E/P ratio. Stability of the dividend yield in the sense of mean reversion implies that, if stock prices are low relative to dividends, prices will increase in the future in order to ensure that the dividend yield returns to its normal historical range. The rule for testing the

dividend yield is similar to the rule based on the E/P ratio: If the dividend yield is above a specified threshold x , market timers are invested in the index. When the ratio drops below the specified threshold, investors exit the index and hold cash.

2.1.3. Book-to-Market Ratio (B/M Ratio)

The third financial ratio used in our experiment is the B/M ratio. Like the two ratios above, the B/M ratio indicates whether the market is underpriced or overpriced. Given an overpriced market reflected by a low B/M ratio, future returns are assumed to be low since the overpricing will be corrected in the future. Thus, as for the E/P ratio and the dividend yield, market timers sell the index and switch to cash when the B/M ratio falls below a certain threshold value x . An increase in the B/M ratio above x triggers the reverse strategy.

2.2. Rules Based on Interest Rates

2.2.1. Short-Term Interest Rate and Discount Rate

The first macroeconomic indicators to be investigated as useful signals for market timing are the short-term interest rate (Treasury bills) and the discount rate announced by the U.S. Federal Reserve. Empirical studies find that falling short-term interest rates are associated with economic expansion and thus predict high stock market returns (e.g., Fama (1981)). Using a simple regression model, Breen et al. (1989) evaluate the ability of treasury bill rates to forecast stock market movements. More recent studies, such as Ang and Bekaert (2007), support the superior forecasting power of short-term interest rates compared to other variables. Prather and Bertin (1997) and Prather and Bertin (1998) test a trading rule based on discount-rate announcements by the U.S. Federal Reserve. Their strategy involves holding stocks when discount rates are decreasing and switching to T-bills when discount rates increase. The trading rule based on both short-term interest rates and discount rates entails exiting the market after the interest rates rise above a certain level x . A drop in the interest rates below the threshold is a signal to switch back to the index.

2.2.2. Long-Term Interest Rate

Since Treasury bonds qualify as a direct alternative to stock investments, the interest rate on such bonds should have even more predictive power for stock returns than the interest rates described in the previous section. If the interest rate increases, we would expect stock prices to go down, and vice versa. The trading rule based on the long-term interest rate is the same as that for the short-term interest rate. A speculator will exit the index and switch to holding cash if the interest rate rises above a certain threshold x ; the reverse strategy is triggered if the interest rate drops below x .

2.2.3. Maturity Spread

Rather than focusing on interest rates themselves, many economists have tried to predict future real economic activity by using the maturity spread, defined as the spread between yields on long-term and short-term bonds. The term structure of interest rates contains useful information about expected real interest rates and expected inflation (e.g., Mishkin (1990b)) and is a useful tool to predict economic growth and thus stock-market returns.¹¹ Intuitively, economic growth and the return on stocks are linked, since stock market returns are expected to be higher during economic expansion. Campbell (1987) and Fama and French (1989), among others, show that the maturity spread is correlated with future stock returns and thus useful for predicting subsequent stock market movements. The theoretical underpinning for the indicator property of the maturity spread, as a proxy for the term structure of interest rates, is given by the expectation theory, according to which the relation between the short-term interest rate and the long-term interest rate depends on the expected short-term interest rate. Tight monetary policy causes economic agents to expect an increase in the short-term interest rate. The term structure of interest rates flattens. Consequently, a change in the maturity spread is associated with a change in the business cycle. In particular, a high maturity spread indicates economic growth and a low spread indicates an economic downturn.

The usefulness of the term structure of interest rates for predicting economic development was, for instance, examined by Estrella and Hardouvelis (1991), Harvey (1997), and Hamilton and Kim (2002). Estrella and Mishkin (1996) apply a probit model to assess the indicator property of the maturity spread and find that it has superior power for predicting recessions compared to other macroeconomic variables. Their model is extended by Resnick and Shoesmith (2002), who find that the maturity spread can be used to forecast downturns in the stock market. Under a rule based on the maturity spread, when the current maturity spread falls below a threshold value x , an investor switches from the index to cash, since the market is expected to yield lower returns than cash during these periods. For values of the spread above the threshold, the investor invests in the index.

2.3. Rules Based on Investor Sentiment

2.3.1. Credit Spread

Sentiment indicators, such as the credit spread, measure the attitude of investors towards the stock market. Fama and French (1989), among others, find the credit spread to be the best predictor for future stock returns, besides the dividend yield and the maturity spread.¹² One widely used sentiment measure is the spread between the yields on BAA-rated and AAA-rated corporate bonds, as applied, for instance, by Fama and French (1989). A decrease in the credit spread indicates lower risk aversion on the part of investors and therefore a good climate for the stock market. An increase in the spread indicates higher risk aversion on the part of investors and therefore a poor climate for the stock market. The rule to be tested is specified as follows: If the credit spread rises above a certain threshold x , speculators switch from the index portfolio to cash holdings; if it falls below x , speculators invest in the index.

2.3.2. Put/Call Ratio

Another implicit measure of investor sentiment is the put/call ratio, defined as the trading volume of puts (or number of puts outstanding) relative to the trading volume of calls (or number of calls outstanding). Often, the ratio is used as a contrarian indicator.¹³ That is,

when the put/call ratio is high (a high number of puts being traded relative to the number of call options being traded), the market is highly pessimistic and a turnaround with an increase in stock prices is expected. On the contrary, a low put/call ratio indicates that the market is overly optimistic and is expected to adjust downwards. Thus, if the ratio falls below a certain threshold x , a sell signal is triggered, while an increase in the ratio above x is a trigger for buying the index.

2.4. Rules Based on Expected Inflation

The negative relation between inflation and stock market returns has been extensively studied. For instance, Fama and Schwert (1977), Fama (1981), Geske and Roll (1983), Stulz (1986), and Kaul (1987) attempt to provide evidence for the stock-return/inflation puzzle.¹⁴ In probably the most widely cited study on the abnormal relationship between expected inflation and stock returns, Fama (1981) claims that the negative relationship simply reflects the fact that anticipated real activity has opposite impacts on stock returns and on expected inflation, but that the relationship is not causal. Extending the work of Fama (1981), Geske and Roll (1983) and Kaul (1987) argue that the correlation between inflation and stock returns is due to monetary policy changes, real economic conditions, and monetization of budget deficits. As mentioned in Section 2, 2.2.3, the maturity spread contains information about the expected inflation rate. In particular, Fama (1990) and Mishkin (1990a), among others, show that the term structures with maturities of more than nine months are useful for predicting expected inflation. We thus rely on the yield spread between the 1-year and the 2-year Treasury bonds as a measure for expected inflation. Under a rule based on expected inflation, investors will switch from being invested in the index to holding cash if the expected inflation rises above a certain threshold x . This may seem nonintuitive, as high inflation would hurt an investor holding cash even more; however, the downturn in the market is expected to be even larger, so that holding cash is the lesser of the two evils.

2.5. Rules Based on Implied Volatility Index

According to conventional wisdom, investors should decrease their stock holdings in periods of high volatility, a negative relation between expected stock returns and volatility having been widely documented. The theoretical underpinning for the risk-return relationship goes back to Merton (1980) and French et al. (1987). The former argues that, under certain conditions, the market risk premium is positively correlated with the variance of the market portfolio. French et al. (1987) find that, if there is an unexpected increase in market volatility, expected volatility is revised upward for future periods. Hence, given that the market risk premium is positively related to the expected volatility of the market portfolio, discount rates will increase and in turn reduce stock prices. Thus, a negative relation between (unexpected) volatility changes and returns is induced. The relation between market volatility and expected returns has been subject to considerable research, such as French et al. (1987), Breen et al. (1989), Campbell and Hentschel (1992), Glosten et al. (1993), and Whitelaw (1994). In the first theoretical study on the relationship between conditional mean independence and the predictability of asset-return signs as well as asset-return volatilities, Christoffersen and Diebold (2006) show analytically that sign predictability follows from volatility dependence. The performance of timing rules based on volatility changes has been investigated, for instance, by Copeland and Copeland (1999), who test the feasibility of market timing based on changes in the implied volatility index, VIX, and find profitable strategies which involve various style and size strategies.¹⁵ Other studies, such as Fleming et al. (2001), Fleming et al. (2003), and Johannes et al. (2002), applying more complex trading strategies than just switching between two asset classes, find that “volatility timing” leads to significant economic benefits. In our paper, under a volatility trading rule, an investor switches to cash if the VIX rises above a certain threshold x . A buy signal is triggered if the volatility index drops below that threshold. All volatility trading rules make use of implied volatility, which serves as a measure of the expected volatility.

2.6. Modified Rules and Combined Indicators

2.6.1. Rules Based on the Bond-Equity Yield Ratio

In addition to the traditional financial ratios, described in Section 2, 2.1, the so-called bond-equity yield ratio (beyr) has become a popular indicator for future stock market movements.¹⁶ Clare et al. (1994), Levin and Wright (1998), Harris and Sanchez-Valle (2000), and Brooks and Persaud (2001), among others, find that this augmented financial ratio is a useful tool for predicting future stock returns and thus a useful instrument for market timing. The bond-equity yield ratio is defined as the ratio of the bond yield to the yield in the stock market, where the dividend yield and the earnings yield (E/P ratio) are commonly used as proxies for the stock market yield. It is assumed that beyr possesses a long-term level which reflects the long-run arbitrage relation between the government bond market and the stock market. If the beyr becomes high relative to its long-term level, equity yields are considered to be low compared to yields on bonds. Stock prices are therefore expected to fall in order to reestablish the long-run equilibrium. In contrast, a low beyr indicates that equities are cheap compared to bonds and that equity yields are expected to decrease in order to restore the long-term relationship. Under a rule based on the beyr, investors invest in the index if the ratio is below a certain threshold x . A rise in the ratio above x is a signal to sell the index and switch to cash.

Shen (2003) employs an indicator closely related to beyr. Instead of using the ratio explained above, Shen (2003) uses the spread between the E/P ratio and interest rates to construct market-timing strategies, finding that investors following such strategies can beat a buy-and-hold strategy.¹⁷ Obviously, the spread contains the same information as the ratio explained above. Hence, we do not include the spreads used by Shen (2003) in our set of market-timing indicators, but rather the ratio of the 3-month Treasury bill rate to the earnings/dividend yield as well as the ratio of the 10-year Treasury bond rate to the earnings/dividend yield.

2.6.2. Rules Based on the Dividend Payout Ratio

Lamont (1998) suggests that, in addition to the dividend yield and the earnings yield, the dividend payout ratio, defined as the ratio of dividends per share to earnings per share, has predictive power for future stock market returns.¹⁸ In particular, he argues that the dividend payout ratio should be positively correlated with future returns, since high dividends typically forecast high returns whereas high earnings typically forecast low returns. Lamont (1998) does not treat earnings and dividends as scaling variables for prices, but rather uses the dividend payout ratio as an explanatory variable for future stock returns. In our study, we identify buy (sell) signals when the dividend payout ratio crosses a threshold x from below (above).

3. Data and Indicator Construction

In our experiment, we apply the market-timing rules to the S&P 500 index and base our analysis on continuously compounded end-of-day returns. Our data for indicator construction are taken from different sources and cover all trading days from January 1, 1980 through December 31, 2007, for a total of 7,305 observations. The year 1980 is used as the observation window in order to compute percentiles and means for first signals. Hence, we compute performance measures of all market-timing rules from the year 1981 onwards.¹⁹

Since data for the put/call ratio and the VIX are not available before 1994, we decided to split the full sample into two subperiods of roughly equal length, 1981-1994 and 1995-2007. Rules based on the put/call ratio and on the VIX are only considered in the latter subperiod. When considering subperiods, the years 1980 and 1994 are used as the observation windows.

The put/call ratio is obtained from Bloomberg. All other time series in our paper are extracted from Thomson Datastream. All time series are at a daily frequency. A detailed description of the applied time series and their codes in the source databases is given in

Appendix A.

For indicator construction, the E/P ratio and the dividend yield can be directly extracted from Thomson Datastream, whereas the B/M ratio needs to be constructed. For the years 1980 through 2005, we calculate the B/M ratio for the S&P 500 by dividing the book value at the end of the previous month by the daily index price series. The book values are calculated from the end-of-month B/M ratios for the S&P 500, which are available on the MSCI Barra webpage.²⁰ For the years 2006 and 2007, we calculated the end-of-year book value for the index from the B/M ratios taken from the S&P webpage.²¹ The daily B/M ratio is then calculated by dividing the book value at the end of the previous year by the daily index price. For indicators based on interest rates, we apply the yields of 3-month and 6-month Treasury bills as well as the discount rate announced by the U.S. Federal Reserve as short-term interest rates. The set of long-term interest rates, which are used as timing indicators, includes yields to maturity of 2-year, 3-year, 5-year, 10-year, and 30-year Treasury bonds. The maturity spread is calculated as the spread between the 3-month Treasury bill yield and the 10-year Treasury bond yield. The yield spread between the 1-year and the 2-year Treasury bonds serves as a proxy for expected inflation. The credit spread is measured as the difference between the yields on a BAA-rated corporate bond and a AAA-rated corporate bond. The bond-equity yield ratio can be calculated in two ways, dividing the yield of the 3-month Treasury bill by the E/P ratio or by the dividend yield. We use both. In addition, we replace the 3-month Treasury bill yield by the 10-year Treasury bond yield when constructing beyr. The dividend payout ratio is defined as the ratio of dividends per share to earnings per share whereas the dividends and the earnings time series can be simply obtained from the dividend yield and the E/P ratio, since the price series is given.

For each indicator in our analysis, we use several fixed numbers as well as the moving average over the preceding 5, 10, 40, and 60 days and the 80th/20th, 90th/10th, and 95th/5th percentiles over the preceding 250 days as threshold values x to obtain switching signals as

described in Sections 2, 2.1 through 2.6.²² For indicators that trigger selling the index when they reach high values, the 80th, 90th, and 95th percentiles are used. The 5th, 10th, and 20th percentiles are used for indicators that trigger switching to cash when they reach low values. When choosing fixed threshold levels for each indicator, we select a fairly large set of parameter values x that lie in the ranges investors and academics may have considered over time. We also consider, for all thresholds, holding a certain position for a period of c days, during which all other signals are ignored. We refer to this period as the waiting period. Alternatively, a time-delay filter is implemented, which requires that a buy/sell signal has to remain valid for d days before a transaction is made. For the waiting period and the time-delay filter, we consider parameter values $c = [1, 2, 5, 10, 20, 30]$ and $d = [1, 2, 5, 10, 20, 30]$. In Appendix B, we provide the parameterizations for the basic set of rules.

In our test, we consider both the average return and the Sharpe ratio as performance measures. In order to compute the Sharpe ratio according to formula (16), we calculate the excess return of a specific rule as the difference between the return from the rule and the risk-free interest rate. As a proxy for the risk-free interest rate we use the Federal funds rate (reported per annum) and convert it into a daily rate according to the formula,

$$r_f^d = \frac{\ln(1 + r_f^a)}{250}, \quad (18)$$

where r_f^d denotes the daily risk-free interest rate and r_f^a denotes the risk-free interest rate per annum. The denominator represents the number of trading days per annum.

4. Empirical Results for Simple Market-Timing Rules

In a first step, we identify the best-performing (simple) market-timing rules for the various sample periods. Table 1 reports the performance of the best performers and the corresponding number of trades under both the mean return criterion (Panel A) and the Sharpe ratio criterion (Panel B). The table also contains the mean return and the Sharpe ratio of the

benchmark. For the full sample, rules based on beyr turn out to be the best performers under both performance criteria, though with different thresholds and different waiting periods. For the first subperiod, exactly the same rule based on beyr is chosen under both performance criteria. For the second subperiod, yet another beyr-based rule is identified as the best rule under the mean return criterion, whereas the best performer under the Sharpe ratio criterion is a rule based on the dividend yield with a threshold $x = 1.75$ and a waiting period $c = 10$. Comparing the returns/Sharpe ratios of the best rules and the benchmark, it can be seen that the buy-and-hold strategy is outperformed under both criteria in all periods. Excess returns of the best rules over the benchmark lie between 2.93% (for the full sample) and 5.57% (for the first subperiod). The number of trades of the best performers varies substantially. The particularly high number of trades of the best rule in the first subperiod might indicate that, when transaction costs are included, other market-timing rules with less frequent trading might be optimal.²³

Table 1 about here

In order to illustrate the performance of the best market-timing rule, Figure 1 plots its cumulative wealth for the full sample from 1981 to 2007, starting with an investment of 100 USD (thick blue line). Figure 1 also includes the cumulative profits of the benchmark (buy-and-hold) strategy (green line) and indicates that the cumulative profit of the best-performing rule clearly dominates that of the benchmark strategy. Even though the return earned by the best rule does not necessarily exceed that of the benchmark at all times, an investor is uniformly better off in terms of cumulative wealth when following the best market-timing rule. Such an investor would have accumulated 2,217 USD over the full sample period, whereas a buy-and-hold investor would have ended up with only 1,081 USD. The cumulative profit of the best performer after one-way transaction costs of 0.05% is indicated by the thin red line. It can be seen that including transaction costs does not qualitatively change the dominance of the best timing rule relative to the benchmark strategy.²⁴

Figure 1 about here

4.1. Test Results under the Mean Return Criterion

Table 2 presents the test results for the performance of the best market-timing rule under the mean return criterion. For each time period, the RC p-values, the SPA p-values, and the corresponding nominal p-values are summarized; in the table, SPA-c refers to the consistent SPA p-value of Hansen (2005) and SPA-l refers to the inconsistent lower bound of the SPA-c p-value.²⁵ The nominal p-values for the full sample and the two subperiods are all well below 0.10, rejecting the null hypothesis of no outperformance of the best rules relative to a buy-and-hold strategy. This result might not be entirely unexpected due to the excess return of those rules relative to the benchmark. For the full sample and the first subperiod, we find the outperformance to be significant at the 5% level; for the second subperiod, the superior performance is significant at the 10% level. In sharp contrast, the RC and the SPA-c p-values are very high for all periods and the null hypothesis is not rejected at any conventional significance level. These findings show that, even though market-timing rules might generate statistically significant excess returns relative to the benchmark when evaluated in isolation, correcting for dependencies between rules leads to insignificant p-values.

Table 2 about here

Figure 2 demonstrates the effects of data snooping and the complexity of dependencies across the set of market-timing rules even more clearly than they can already be seen in Table 2. In Figure 2, we plot the number of the model against its average annualized return, measured on the left y-axis and identified by the scattered points. Also shown is the sequentially updated maximum average return (thin red line) from a certain number of models (indicated on the x-axis), the corresponding RC p-value (measured on the right y-axis and represented by the solid blue line), and, for comparison, the benchmark return (horizontal green line).²⁶ For the final assessment, only the numbers on the far right of the graph (the

terminal maximum mean return and the terminal RC p-value) matter, since the order of models is arbitrary. Nevertheless, the figure gives a fascinating picture of how the maximum mean return and the corresponding RC p-value evolve when additional models are sequentially included. The maximum mean return increases quickly to approximately 11%, well above the benchmark return, but rises no further after the first 3,900 trading rules have been evaluated. The p-value, however, does not remain constant, but increases gradually up to 0.74 after the same number of models have been considered. It can be seen that, while the marginal trading rules do not improve on the maximum mean return, the data-snooping adjusted p-value gradually increases, since the effective span of market-timing rules widens. After superior market-timing rules are included around model 3,900, the RC p-value drops to 0.52. From this point until all 6,792 rules are considered, the pattern is similar to that already described. Since no improvements occur in the maximum performance statistic as the number of rules increases, the RC p-value is subject to the same “dilution effect” as before and increases to a terminal value of 0.60.

Figure 2 about here

4.2. Test Results under the Sharpe Ratio Criterion

The test results for the best-performing rule under the Sharpe ratio criterion are summarized in Table 3, which reports the RC and the SPA p-values for all sample periods, as well as the corresponding nominal p-values. The results are qualitatively the same as those under the mean return criterion. Nominal p-values close to zero for the full sample period and the first subperiod suggest that, without taking data snooping into account, the best rules significantly beat a buy-and-hold strategy at the 5% significance level. The highest nominal p-value for the second subperiod, 0.09, is still significant at the 10% level. For all periods, data-snooping adjusted p-values are even higher than under the mean return criterion, indicating that the superior performance of the best rules is insignificant for the full sample and for both subperiods. Thus, accounting for dependencies between market-timing rules renders the significant performance of the best-performing rule (when considered in isolation) insignif-

icant. Once more, our findings illustrate the effect of data snooping on statistical inference and emphasize the importance of appropriate methods to correct for data-snooping biases.

Table 3 about here

As before, we graphically illustrate the evolution of the economic and statistical performance of the best-performing rule under the Sharpe ratio criterion for the full sample period. As can be seen from Figure 3, the maximum Sharpe ratio increases to a value of 0.35 after including 35 models, which decreases the RC p-value to 0.73. Including an additional 300 rules—among them several models that remarkably underperform the benchmark—leads to a sharp increase in the RC p-value to 0.96. Over the next 1,200 models, the maximum performance statistic does not change and the RC p-value rises only slightly. At around model 1,500, we find an improvement in the maximum performance statistic, associated with a drop in the RC p-value. Over the remaining rules, the maximum Sharpe ratio improves only slightly to roughly 0.46. Since the improvement is small as the set of rules becomes larger, we observe that the RC p-value increases gradually to a highly insignificant terminal value of 0.98, displaying the effects of data snooping.

Figure 3 about here

In summary, considering simple market-timing rules, each of which is based on one indicator only, we find that some of them, when considered in isolation, produce superior performance relative to a buy-and-hold strategy. This finding corresponds to the results of the market-timing literature. However, once we account for dependencies across rules—that is, once we cease to ignore the fact that the best rules are drawn from a large set of market-timing rules—we no longer observe that superiority.

5. Empirical Results for Complex Market-Timing Rules

So far, our set of market-timing rules includes only simple rules, each of which relies on the information from only one indicator. In practice, investment decisions are likely to be based on a broader information set and thus on complex trading strategies that incorporate information from various indicators. We therefore add to our basic set of simple rules two groups of complex trading strategies: learning rules and voting rules. As before, all rules rely only on historically available information and may therefore be implemented in practice. For all complex strategies, we consider, as in previous sections, holding cash as the exit position when the rule indicates selling the index.

Under a learning rule, an investor, at certain times, determines the best-performing rule among all simple trading rules over the last e days, the evaluation period, and then follows the signal of this particular rule. The performance of each simple rule is evaluated based on two measures: the sum of daily returns and the average (daily) log return, both over the preceding e days. In addition to the evaluation period e , we consider a review span r , which determines how often the evaluation process for the best-performing rule is conducted and a possible switching is initiated. This leaves us with a total of 1,482 learning rules.

Voting rules are defined as follows: Each simple rule in our universe has one vote, either to invest in the index (pro index) or to exit the market (contra index), according to that rule's switching signal. The signal for a voting rule is based on the ratio of pro index and contra index votes. Traders sell the index if b percent of all votes signal exiting the market, and vice versa. As with the learning rules, we implement a review span r , for a total of 494 voting rules. Parameterizations for all complex timing rules are given in Appendix C.

As with the basic set of simple rules, we first identify the best market-timing rules among all the rules in the extended set. In Table 4, Panel A, we report their mean returns and number of trades for all sample periods and, for comparison, the mean return of the

benchmark. Panel B presents the Sharpe ratios of the best rules and of the benchmark, as well as the number of trades triggered by the best market-timing rules. We find that for each period the same rules are selected as best-performers under both performance criteria. Interestingly, for all periods, learning rules turn out to perform best in the extended set of rules. Hence, the maximum performance for all periods is higher than in the previous exercise, in which we considered only simple rules. We find the largest increase in the maximum performance for the first subperiod with a rise of the mean return from 14.59% (for the best rule in the set of simple rules) to 24.24%, which equals an outperformance over the benchmark of more than 15% per annum. This remarkable excess return of the best-performing rule relative to the benchmark gives rise to the conjecture that this outperformance may be statistically significant, even with adjustment for data-snooping biases. Since the number of trades of the best learning rules are even higher than for the best rules from the set of simple rules, the robustness check on the impact of transaction costs in Section 6, 6.2 is even more valuable.

Table 4 about here

5.1. Test Results under the Mean Return Criterion

In order to assess the outperformance of the best-performing rule over a buy-and-hold strategy, we run the RC and the SPA for the extended set of market-timing rules, which includes complex strategies as well as simple rules. Table 5 presents nominal p-values and data-snooping adjusted RC and SPA p-values for each period based on the mean return criterion. As before, SPA-c denotes the consistent p-value of Hansen (2005)'s SPA and SPA-l is the lower bound of the consistent SPA p-value. Not surprisingly, due to the outperformance of the best rules over the benchmark, nominal p-values are significant at the 5% level (at least) for all periods. However, data-snooping corrected RC and SPA p-values are no longer significant at any conventional significance levels for the full sample and the second subperiod. That is, the best-performing rules, which showed significant outperformance over the benchmark when considered in isolation, no longer beat the buy-and-hold strategy once we correct for dependencies across all rules. Interestingly, according to the data-snooping

adjusted RC and SPA p-values, the null hypothesis that market timing does not produce superior returns can still be rejected at the 10% level for the first subperiod. This result supports our previous conjecture that the best rule for the subperiod from 1981 to 1995 may still significantly beat the benchmark, even after data-snooping correction.

Table 5 about here

In Figure 4, we plot the evolution of the maximum mean return and the RC p-value across all rules from the extended set for the first subperiod (1981-1994), in which we found statistically significant outperformance of the best rule even after we corrected for data-snooping biases. The RC p-value terminates at 0.05, indicating that, even after adjustment for data snooping is made, the optimal rule does indeed contain valuable economic information. The graph of the maximum mean return for this subperiod shows that, after including only 67 models, the maximum performance statistic increases to 9.41% and thus above the return of the buy-and-hold strategy. Including roughly 1,400 additional rules increases the maximum performance statistic to 12.23% without rendering the RC p-value significant. Since from this point onwards the improvement of the maximum performance is small and the effective span of market-timing rules becomes larger, the p-value increases gradually to a value of 0.58 after around 6,100 models, displaying the effects of data snooping. Adding additional rules gradually increases the maximum mean return to approximately 25% after 6,793 models are inspected and causes a decline in the RC p-value to a significant terminal value of 0.05. This is even more interesting, given that only very few rules show remarkably high average returns, while the majority clearly underperform the benchmark. The performance of the best rule is apparently so strong that, even after data-snooping correction, it leads to a significant RC p-value.

Figure 4 about here

5.2. Test Results under the Sharpe Ratio Criterion

Table 6 presents the test results for the best-performing rule, drawn from the extended set, under the Sharpe ratio (SR) criterion. We report the nominal p-values as well as the data-snooping adjusted RC and SPA p-values for all sample periods and denote the consistent p-value of Hansen (2005)’s SPA with SPA-c and the lower bound of the consistent SPA p-value with SPA-l. The results are very similar to those under the mean return criterion. Nominal p-values are zero or close to zero for all sample periods, suggesting that the best-performing rules significantly outperform a buy-and-hold strategy at the 1% level when data-snooping biases are ignored. As indicated by insignificant RC and SPA p-values for the full sample and the second subperiod, the best rules for these periods no longer significantly beat a buy-and-hold strategy when adjustment for data snooping is made. However, due to the remarkable outperformance of the best rule ($SR = 1.72$) over the benchmark ($SR = 0.12$) in the first subperiod, RC and SPA p-values remain significant at the 10% level, even after correcting for data snooping.

Table 6 about here

Figure 5 plots, for the first subperiod (1981-1994), the evolution of the maximum Sharpe ratio and the RC p-value across all rules from the extended set. As before, we present the graph for this subperiod since even data-snooping adjustments do not render the outperformance of the best-performing rule insignificant. Including about 130 models increases the maximum SR above the SR of the benchmark and leads to a decline in the RC p-value to slightly less than 0.80. From there until model 6,792, the maximum performance statistic only improves a little and the RC p-value fluctuates but never becomes significant. Including the best performing market-timing rule (model 6,793) leads to a tremendous drop in the RC p-value from 0.87 to a significant value of 0.05. After model 6,793 the majority of rules underperform the buy-and-hold strategy, as under the mean return criterion, and very few rules show an SR significantly above that of the benchmark. However, the outperformance

of the best-performing rule is strong enough to lead to a significant terminal RC p-value.

Table 5 about here

Overall, the results for the extended set of rules are very similar to those for the basic set. When no adjustment for data snooping is made, certain rules significantly outperform a buy-and-hold strategy. In general, however, data-snooping adjustments render the superior performance of the best rules insignificant. Only for the first subperiod (1981-1994) does the outperformance of the best-performing rule remain significant after we correct for data-snooping biases.

6. Robustness Checks

We perform a series of robustness checks for both the set of simple rules and the extended set of rules. Since the results under the two performance criteria are very similar in previous sections, we restrict all robustness checks to experiments based on the mean return criterion.

6.1. Impact of Relaxing the Short-Selling Constraint

In our first robustness check, we investigate the performance of market-timing rules when the short-selling constraint is relaxed. Thus far, we assumed that an investor receiving sell signals exits the market and switches to cash. However, instead of holding cash after selling the index, investors may want to short sell the index in order to gain from falling prices. Short selling the index has been facilitated by the emergence of basket securities, such as index futures and exchange-traded funds replicating indices, as they allow for inexpensive index tracking and for a potential short selling of the index. Consequently, short selling the index might realistically be considered by investors in practice and might change the results found in previous sections. On the one hand, if predictions from the indicators are correct—if market downturns are predicted successfully—short selling the index when exiting the market

will yield superior returns relative to holding cash. The performance of market-timing rules would thus be higher than in previous sections and the test results of the superior performance of the best market-timing rules might become more significant. On the other hand, if market-timing strategies fail, investors who short sell the index instead of holding cash when indicators signal to exit the market will face higher losses. Test results of the superior performance of market-timing rules would consequently tend to be less significant.

In order to determine whether being short the index during predicted market downturns improves on the maximum return for the set of market-timing rules and thus on the significance of superior performance, we run the RC and the SPA while changing the values of the timing function in equation (2) to 1 for “invest in the stock market” and -1 for “short sell the index.” Table 7 reports nominal p-values as well as RC and SPA p-values for the basic set of rules (Panel A) and the extended set of rules (Panel B). Nominal p-values in Panel A indicate that, considered in isolation, the best-performing of the simple rules outperforms the benchmark at the 5% significance level in the full sample and in the first subperiod and at the 10% level in the second subperiod. After correcting for data snooping, the outperformance of the best market-timing rule is not significant in any of the periods, as indicated by high RC and SPA p-values. Furthermore, we find that, while nominal p-values are very similar to those in Section 4, 4.1, Table 2, where holding cash was used as the exit strategy, data-snooping corrected p-values show a relatively larger change. For both the first and second subperiods the RC and SPA-c p-values decrease, while for the full sample these p-values increase.

For the extended set of market-timing rules, we find that, in general, nominal p-values and data-snooping adjusted p-values change marginally relative to the results in Section 5, 5.1, Table 5. Only for the second subperiod do the RC and SPA-c p-values increase substantially. The significance of the results, however, remains the same; nominal p-values for the full sample and for both subperiods are low and significant. High data-snooping corrected

p-values for the full sample and the second subperiod indicate that the outperformance of the best market-timing rule is not statistically significant after data-snooping adjustments are made. As in Section 5, 5.1, RC and SPA p-values for the first subperiod are significant; correcting for data snooping does not render the outperformance of the best rule over the buy-and-hold strategy insignificant.

In summary, we find that, for both the basic set and the extended set of rules, allowing for short selling the index does not qualitatively change the test results obtained in previous sections.

6.2. Impact of Transaction Costs

Thus far, we have not considered the effects of transaction costs on the profitability of market-timing rules and thus on our test results. The impact might indeed be significant, particularly because various market-timing rules involve frequent switching between the index and cash. As a result of frequent trading, the outperformance of timing rules relative to the benchmark is lower. Furthermore, it may happen that rules with a lower trading frequency turn out to be the best performers once transaction costs are taken into account. Therefore, we perform the same exercises as before, but incorporate transaction costs in our analysis. Since a historical series for transaction costs is not available, and since the amount of those costs would vary among different types of investors, we follow Fama and Blume (1966) and use transaction costs equal to 0.05% per one-way trade.²⁷

Table 8 presents the test results for the performance of the best market-timing rule in both the basic set of rules (Panel A) and the extended set of rules (Panel B) when considering transaction costs of 0.05% per one-way trade. Not surprisingly, for both sets of rules, all p-values are higher in the presence of transaction costs. For the set of simple rules, nominal p-values indicate that the outperformance of the best rule, when considered in isolation, is significant, even when transaction costs are included. We observe significance at the 5% level

for the full sample and the first subperiod and at the 10% level for the second subperiod. However, as in the case without transaction costs, the null hypothesis of no superior performance of the best rule cannot be rejected at conventional significance levels for any period, once data-snooping adjustments are made.

For the extended set of rules, the test results are quite similar to those without transaction costs. Nominal p-values are significant for the full sample and for both subperiods. However, data-snooping corrected p-values for the full sample and the second subperiod are insignificant. For the first subperiod (1981-1994), the null hypothesis of no superior performance of the best rule is rejected at the 10% level.

Thus, including transaction costs does not lead to any qualitative change in the results for either the basic set of rules or the extended set of rules.

6.3. Sensitivity to Changes in the Smoothing Parameter

In this section, we examine whether our test results are sensitive to the choice of the smoothing parameter q . The Politis and Romano (1994) bootstrap relies on the parameter q to determine the length of the blocks ($1/q$) resampled from the observed data. A smaller parameter value of q corresponds to a longer block length and is appropriate for data with more dependence, and vice versa.

Thus far, we have chosen a smoothing parameter of 0.5 in all experiments. We now consider parameter values of 0.5, 0.1, and 0.01, corresponding to block lengths of 2, 10, and 100. Table 9 provides RC and SPA p-values along with the corresponding nominal p-values for all sample periods and for the three different values of the smoothing parameter. In order to conserve space, we show the results of this robustness check only for the basic set of rules.²⁸ It can be seen that, for all subperiods, the nominal p-values deviate marginally for different choices of q , the maximum deviation amounting to 0.05 for the second subperiod and a

change in the smoothing parameter from $q = 0.5$ to $q = 0.01$. The only qualitative change is found for precisely that subperiod, where the nominal p-value turns insignificant. Similarly, the data-snooping adjusted RC and SPA-c p-values for all periods fluctuate marginally. The highest deviation is found for the RC p-value in the first subperiod. With a change in the smoothing parameter from $q = 0.5$ to $q = 0.01$, the RC p-value decreases from 0.275 to 0.168. In none of the specifications do we find a qualitative change in the data-snooping adjusted p-values. Overall, we find that our results are robust, not sensitive, to the choice of the smoothing parameter q .

7. Conclusion

This paper examines the profitability of market-timing rules drawn from a comprehensive universe of rules, while accounting for the potential effects of data snooping. Examining a huge set of rules on the same data set bears the risk of biased statistical inference due to data snooping. That is, potential superior performance of some rules over a benchmark may not be due to any genuine merit in the model (rule), but rather to pure chance. We address this concern by applying appropriate techniques to correct for data-snooping biases. Precisely, we use White (2000)’s “Reality Check” (RC) and Hansen (2005)’s test for superior predictive ability (SPA) in order to quantify such data-snooping biases and to investigate the superior performance of the best market-timing rule over a buy-and-hold strategy, while accounting for dependencies across all rules.

First, we perform the tests for a basic set of simple rules, each of which is based on a single indicator only, and find that, when considered in isolation, the best-performing rule beats a buy-and-hold strategy over the full sample period from 1981 to 2007 and over both subperiods. Applying the RC and SPA tests to adjust for the effects of data snooping, we do not find superior performance of the best market-timing rule over the benchmark in any period. These results emphasize the importance of using appropriate methods to correct for

data-snooping biases.

Second, we extend the set of market-timing rules to include complex strategies which are based on information from several indicators. As before, we find significant superior performance of the best rules when dependencies across rules are ignored. But when considered in the context of the full universe of rules, the best-performing rules do not significantly outperform the benchmark in either the full sample or the subperiod from 1995 to 2007. Interestingly, the outperformance of the best rule in the subperiod from 1981 to 1994 remains significant even after data-snooping adjustment is made.

We perform a series of robustness checks and find that our results are not qualitatively changed by implementing transaction costs, allowing for short selling the index, or changing the smoothing parameter which determines the block length of the bootstrap used in our tests.

Appendix

Appendix A. Data Description

Table 10 summarizes the data used in our experiment covering the period from January 1, 1980 to December 31, 2007. The format of the table is *name* and *code*, as used in the source databases (Thomson Datastream and Bloomberg), as well as a brief *description* of each time series.

Table 10 about here

Appendix B. Parameterization for Simple Rules

Table 11 presents parameter values/thresholds x for each indicator used in our experiment. As described in the text, the moving averages over 5, 10, 20, and 60 days and the 80th/20th, 90th/10th, and 95th/5th percentiles over the preceding 250 days are used as (standard) threshold values. For indicators that trigger selling the index when reaching high values, the 80th, 90th, and 95th percentiles are used. The 5th, 10th, and 20th percentiles are used for indicators that trigger switching to cash by reaching low values. In Table 11, we list only the additional threshold values x specific for each rule. We use the notation $x = [lowest\ value : i : highest\ value]$ to indicate that threshold values increase from *lowest value* to *highest value* in increments of i . For each indicator, we consider holding a certain position for a period of c days (waiting period), during which all other signals are ignored. Alternatively, a time-delay filter is implemented, which requires that a buy/sell signal has to remain valid for d days before a transaction is made. In our experiment, we consider parameter values $c = [1, 2, 5, 10, 20, 30]$ and $d = [1, 2, 5, 10, 20, 30]$.

Table 11 about here

Appendix C. Parameterization for Complex Rules

Learning Rules

The following parameter values for the evaluation period e and the review span r of the learning rules are used:

$$e = [1 : 5 : 120, 130 : 10 : 250]^{29} \text{ and}$$

$$r = [1 : 5 : 120, 130 : 10 : 250].$$

Voting Rules

The following parameter values for the review span r and the proportion interval b of the voting rules are used:

$$r = [1 : 5 : 120, 130 : 10 : 250] \text{ and}$$

$$b = [0.2 : 0.05 : 0.8].$$

Notes

¹The set of indicators to time the market can be found in Section 2, where we specify the market-timing rules for our experiment. An overview of relevant literature on each indicator is given in the corresponding subsections.

²Throughout the paper, we use a simple buy-and-hold strategy as the benchmark.

³The dangers of data snooping are highlighted, for instance, in Lo and MacKinlay (1990), Sullivan et al. (1999), Ferson et al. (2003), and Ang and Bekaert (2007).

⁴From here on, we refer to these sets of rules as the basic set and the extended set.

⁵However, we do not consider market-timing strategies that rely on sophisticated models rather than on simple indicators. For instance, we do not include regression-based market-timing strategies as in Breen et al. (1989) and Fuller and Kling (1994).

⁶For the subperiod from 1995 to 2007, the sets of market-timing rules are even larger, since for this time period we consider additional indicators based on data availability. Further details are found in Section 2.

⁷We first outline the procedure for using raw returns, but we will also use the Sharpe ratio (SR) as a performance measure.

⁸The choice of the block length is discussed in more detail in Section 6, 6.3.

⁹Results of a sensitivity analysis for changes in the smoothing parameter are shown in Section 6, 6.3.

¹⁰Sullivan et al. (2001), p. 259, also discuss the issue of including too many “irrelevant” rules, but argue that a loss of power is rather unlikely.

¹¹Since there is a separate line of research on the relationship between expected inflation and stock returns, we discuss the indicator property of the expected inflation component in the term structure of interest rates separately in Section 2, 2.4.

¹²Using sentiment indicators to predict future stock market movements falls into the literature which tries to link business conditions and expected stock market returns (e.g., Campbell and Diebold (2008)).

¹³Even though Fisher and Statman (2000) do not explicitly consider the put/call ratio, they find that sentiment indicators should trigger contrarian investment strategies.

¹⁴The observed negative relationship between stock market returns and different measures of expected and unexpected inflation appears to contradict the so-called Fisher Hypothesis (Fisher (1930)), according to which there should be a positive relationship between the expected inflation and nominal asset returns; claims on real assets, such as stocks, should provide hedging against inflation.

¹⁵The VIX is constructed to represent the implied volatility of a synthetic at-the-money option contract on the S&P 500 index with a maturity of 30 days. It is constructed from eight S&P 500 index options and uses a dividend-adjusted Black-Scholes model.

¹⁶In the literature, the ratio is sometimes referred to as the gilt-equity yield ratio (geyr). See, for instance, Clare et al. (1994).

¹⁷Pesaran and Timmermann (1995) employ the spread between the E/P ratio and interest rates as an explanatory variable for future stock market returns, but do not explicitly assess its usefulness for market-timing strategies.

¹⁸However, Lettau and Ludvigson (2001), using a different data set, find the payout ratio to be statistically insignificant.

¹⁹Throughout the paper, a time period from year A to year B starts on the first trading day of year A and ends on the last trading day of year B.

²⁰End-of-month B/M ratios for the S&P 500 for the years 1980 through 2005 are available at <http://www.barra.com/Research/Fundamentals.aspx>.

²¹End-of-year B/M ratios for the S&P 500 for the years 2006 and 2007 are available at http://www2.standardandpoors.com/portal/site/sp/en/us/page.topic/indices_500/2,3,2,2,0,0,0,0,0,5,7,0,0,0,6,0.html.

²²Henceforth, moving averages and percentiles are referred to as standard thresholds.

²³In Section 6, 6.2, we check for the robustness of the results when transaction costs are included.

²⁴A more detailed robustness check with respect to transaction costs is performed in Section 6, 6.2.

²⁵In the following, we base our assessments on the RC p-value and the consistent SPA-c p-value. The inconsistent lower bound for the consistent SPA p-value, SPA-l, is reported for information purposes.

²⁶Since the RC and the SPA provide similar results, we restrict all graphical illustrations to p-values obtained from the RC.

²⁷We have also tried transaction costs of 0.02% per one-way trade, taking into account that transaction costs might have been lower during our sample period. The results are very similar and the conclusions remain the same.

²⁸We also performed this robustness check for the extended set of rules. The results are similar and the conclusions remain the same.

²⁹We use this notation to indicate that parameter values increase from 1 day to 120 days in increments of 5 days and from 130 days to 250 days in increments of 10 days. The same notation is used analogously for parameter values for the evaluation period and the proportion interval.

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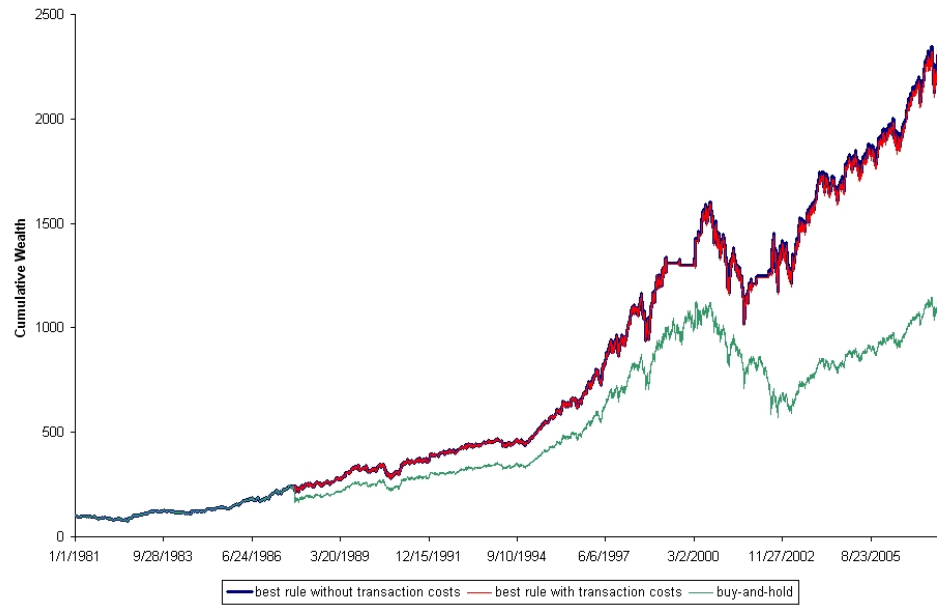


Figure 1: Cumulative Wealth of the Best Market-Timing Rule and the Buy-and-Hold Strategy over the Full Sample

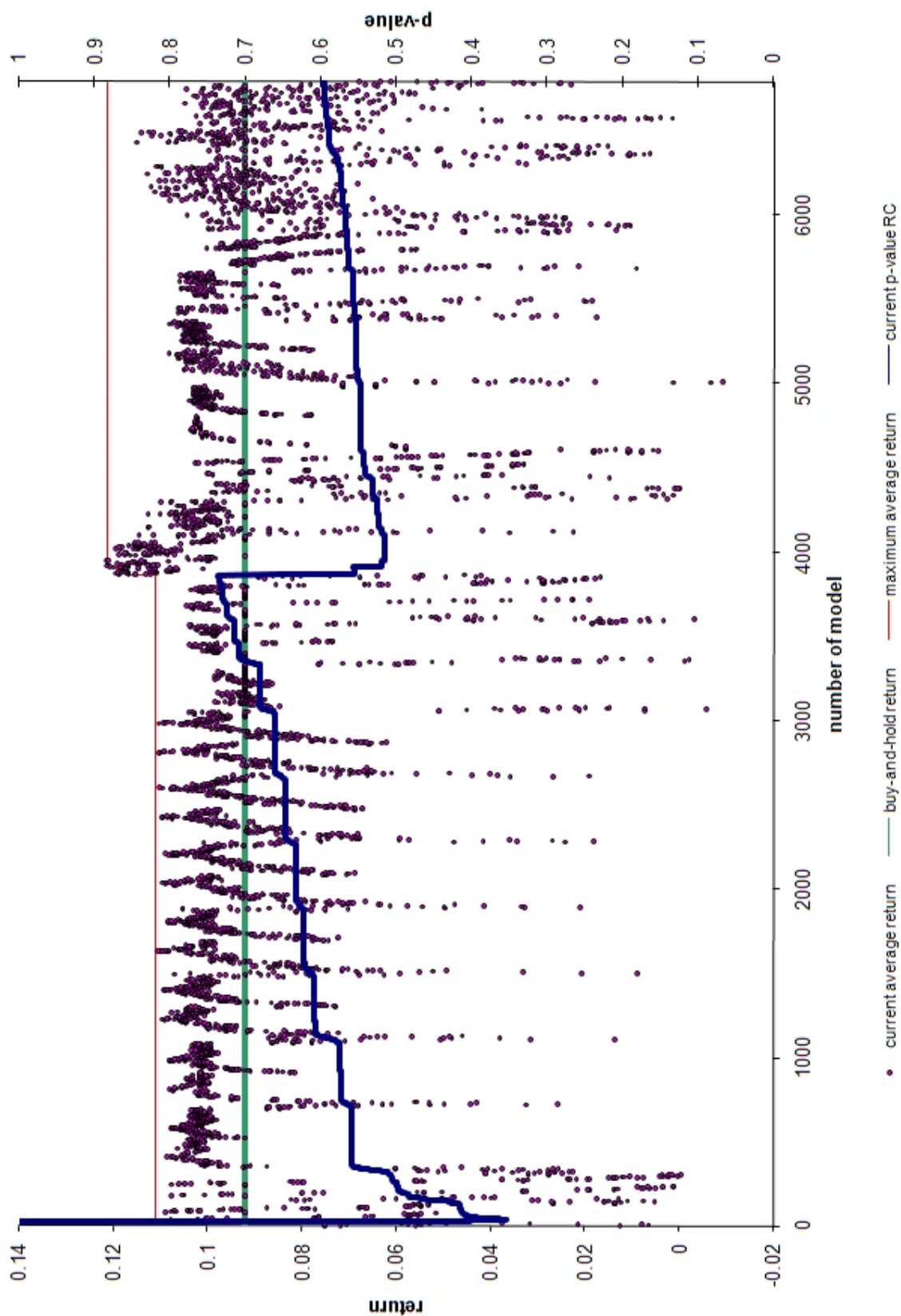


Figure 2: Economic and Statistical Performance of the Best Market-Timing Rule in the Basic Set under the Mean Return Criterion: Full Sample (1981-2007). For each trading rule k , indexed on the x-axis, the annualized average return over the full sample (1981-2007) is identified by the scattered points. In addition, the current maximum mean return in the basic set of market-timing rules (thin red line), the associated current “Reality Check” p-value (solid blue line), and the benchmark return (horizontal green line) are plotted.

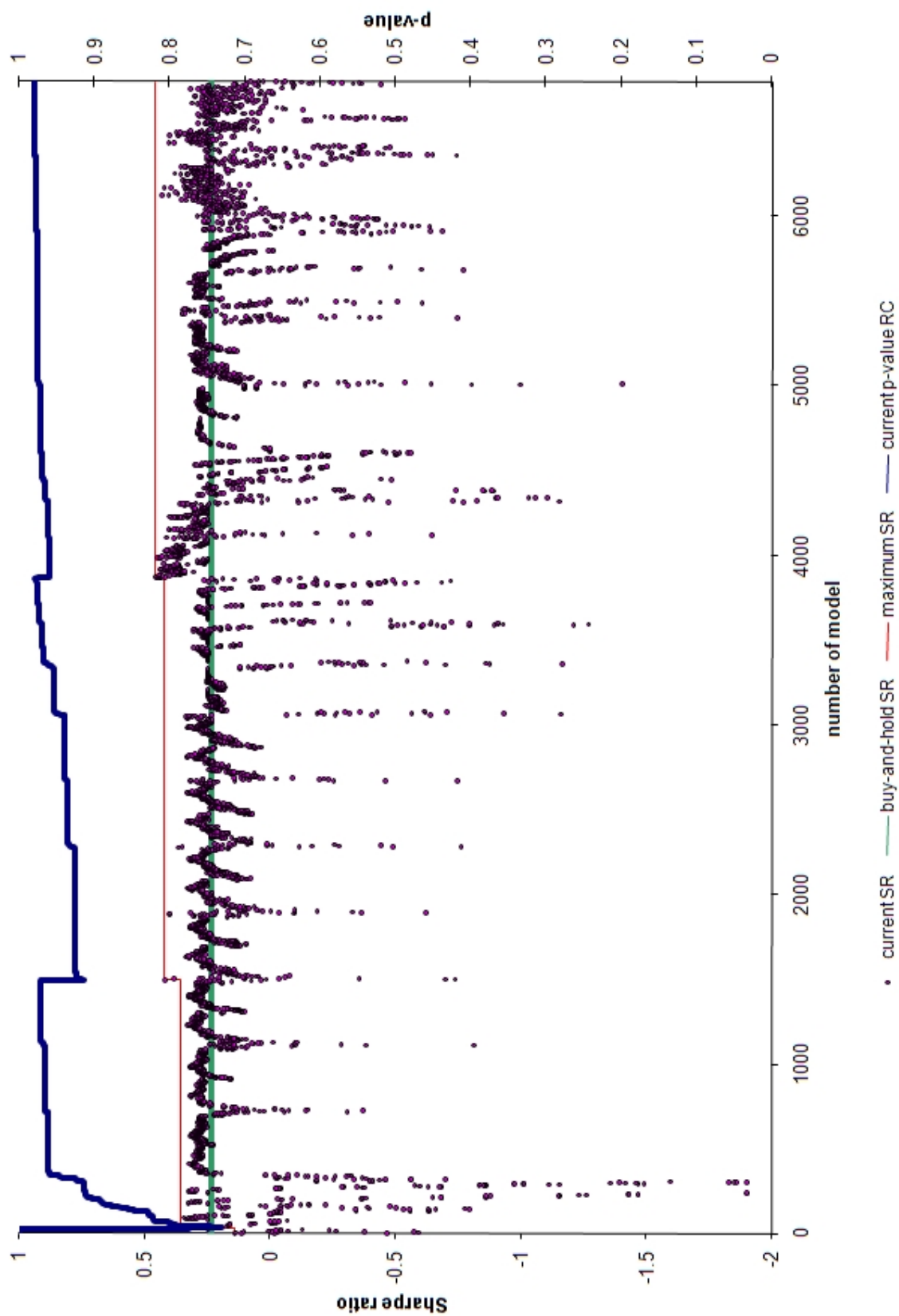


Figure 3: Economic and Statistical Performance of the Best Market-Timing Rule in the Basic Set under the Sharpe Ratio Criterion: Full Sample (1981-2007). For each trading rule k , indexed on the x-axis, the annualized Sharpe ratio over the full sample (1981-2007) is identified by the scattered points. In addition, the current maximum Sharpe ratio in the basic set of market-timing rules (thin red line), the associated current “Reality Check” p-value (solid blue line), and the benchmark Sharpe ratio (horizontal green line) are plotted.

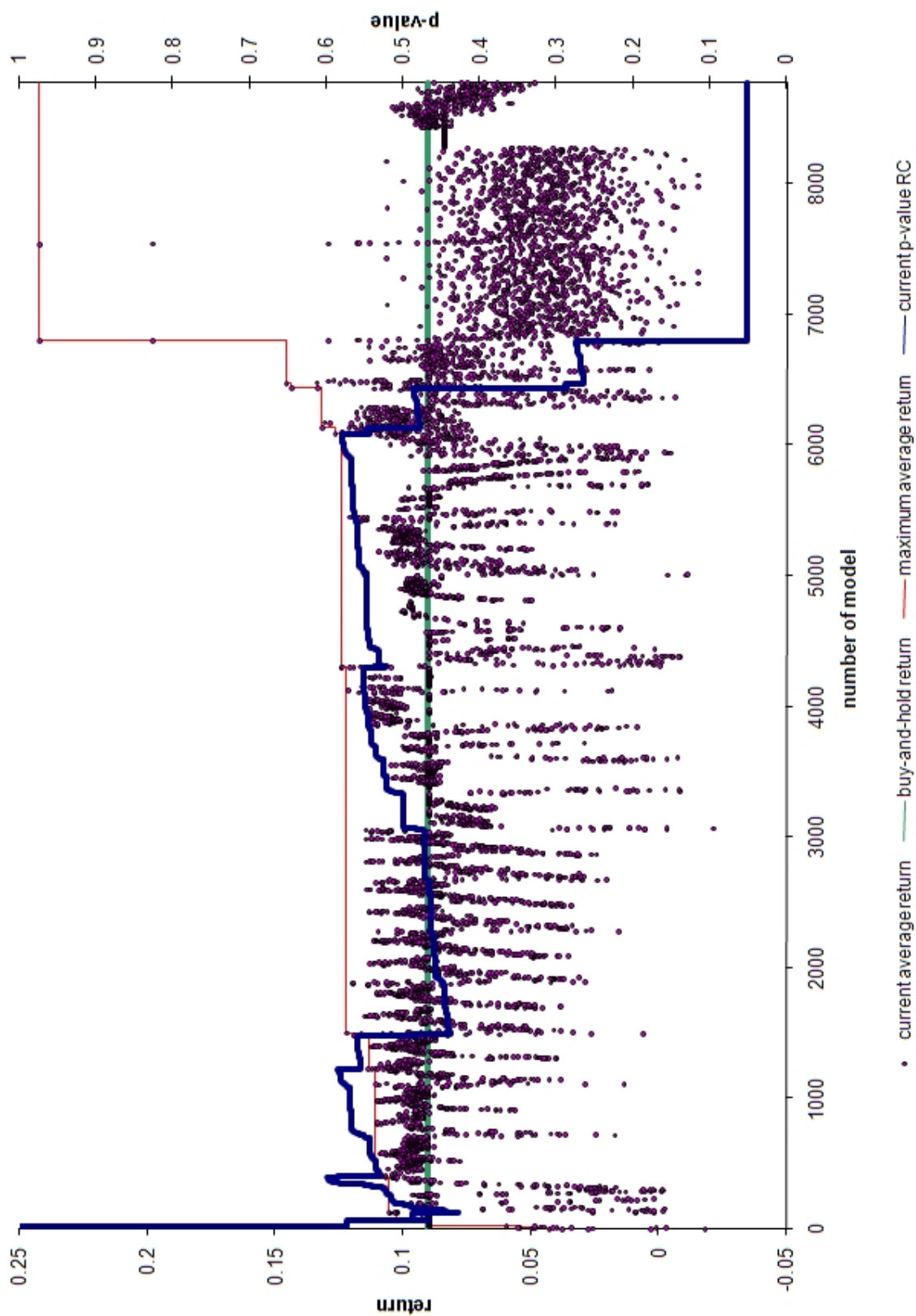


Figure 4: Economic and Statistical Performance of the Best Market-Timing Rule in the Extended Set under the Mean Return Criterion: Subperiod I (1981-1994). For each trading rule k , indexed on the x-axis, the annualized average return over the first subperiod (1981-1994) is identified by the scattered points. In addition, the current maximum mean return in the extended set of market-timing rules (thin red line), the associated current “Reality Check” p-value (solid blue line), and the benchmark return (horizontal green line) are plotted.

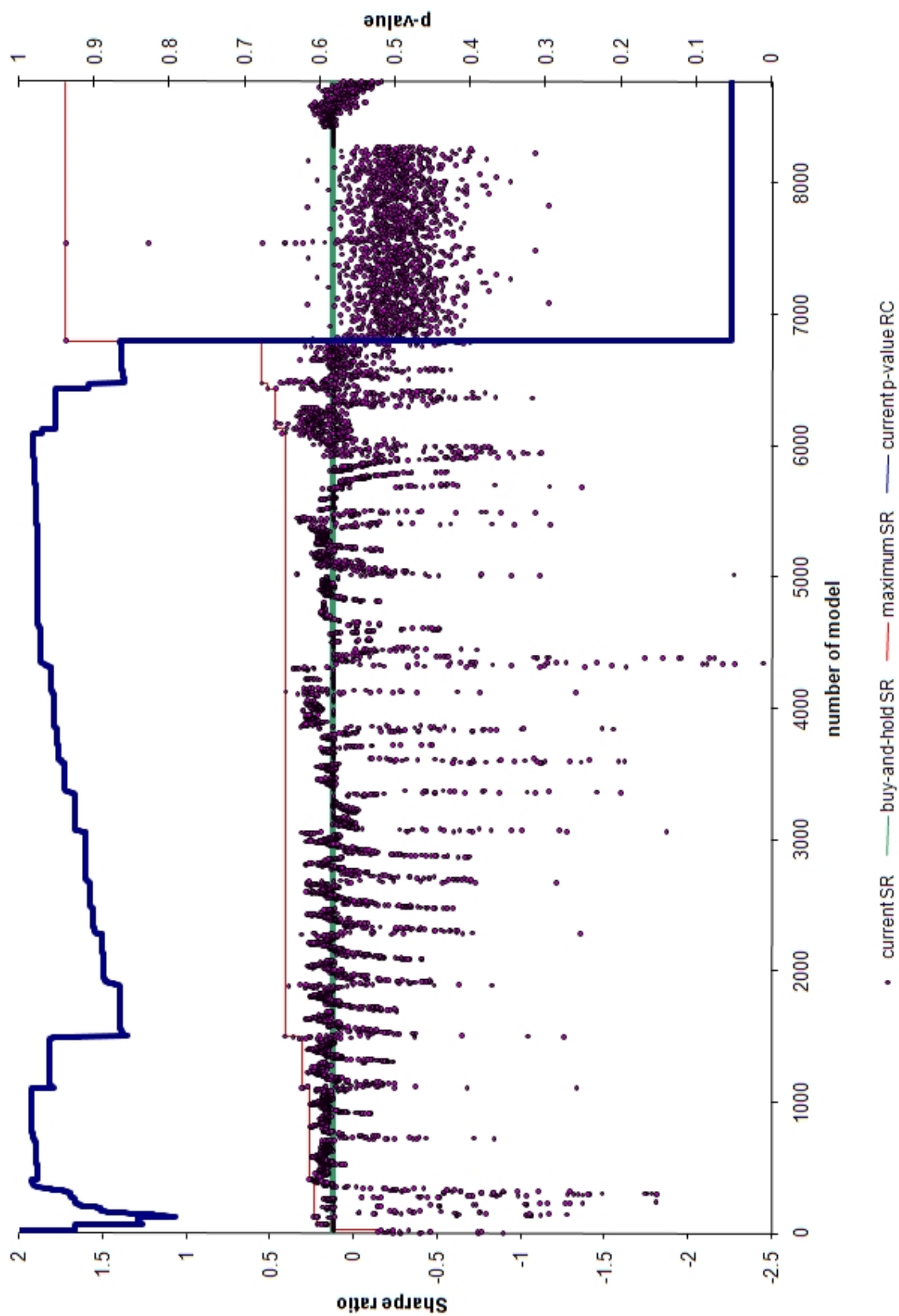


Figure 5: Economic and Statistical Performance of the Best Market-Timing Rule in the Extended Set under the Sharpe Ratio Criterion: Subperiod I (1981-1994). For each trading rule k , the annualized Sharpe ratio over the first subperiod (1981-1994) is identified by the scattered points. In addition, the current maximum Sharpe ratio in the extended set of market-timing rules (thin red line), the associated current “Reality Check” p-value (solid blue line), and the benchmark Sharpe ratio (horizontal green line) are plotted.

Table 1: Best Market-Timing Rules in the Basic Set under the Mean Return Criterion and the Sharpe Ratio Criterion

Panel A Sample	Best Trading Rule	Mean Return Criterion		
		# Trades	Model Return	Benchmark Return
Full Sample (1981 - 2007)	Beyr-EP threshold 1.9, $c = 20$	20	12.15%	9.22%
Subperiod I (1981 - 1994)	Beyr-DY mean (20 lags), $d = 5$	244	14.59%	9.02%
Subperiod II (1995 - 2007)	Beyr-EP threshold 1.6, $c = 5$	22	12.70%	9.35%

Panel B Sample	Best Trading Rule	Sharpe Ratio Criterion		
		# Trades	Model SR	Benchmark SR
Full Sample (1981 - 2007)	Beyr-EP threshold 1.55, $c = 5$	58	0.457	0.237
Subperiod I (1981 - 1994)	Beyr-DY mean (20 lags), $d = 5$	244	0.582	0.122
Subperiod II (1995 - 2007)	DY threshold 1.75, $c = 10$	20	0.683	0.350

This table reports the performance of the best market-timing rule under the mean return criterion (Panel A) and the Sharpe ratio criterion (Panel B) for each sample period. The Sharpe ratio is denoted by SR. The table also contains the mean return and the Sharpe ratio of the buy-and-hold strategy (benchmark model). The mean returns and the Sharpe ratios are annualized. Lower case letters d and c denote the length of the time-delay filter and of the waiting period, respectively.

Table 2: Performance of the Best Market-Timing Rules in the Basic Set under the Mean Return Criterion

Sample	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.030	0.598	0.030	0.569	0.361
Subperiod I (1981 - 1994)	0.035	0.275	0.035	0.257	0.169
Subperiod II (1995 - 2007)	0.073	0.885	0.073	0.884	0.742

This table presents the performance of the best market-timing rule under the mean return criterion. For each period, the table shows the “Reality Check” (RC) and the SPA p-values along with the corresponding nominal p-values. SPA-c refers to the consistent p-value of Hansen (2005)’s SPA; SPA-l is the lower bound of the consistent SPA p-value. The nominal p-values are obtained by applying the RC and the SPA testing procedures to the best market-timing rule only, without correcting for data-snooping biases.

Table 3: Performance of the Best Market-Timing Rules in the Basic Set under the Sharpe Ratio Criterion

Sample	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.027	0.981	0.027	0.917	0.628
Subperiod I (1981 - 1994)	0.014	0.867	0.014	0.685	0.316
Subperiod II (1995 - 2007)	0.088	0.933	0.088	0.914	0.750

This table presents the performance of the best market-timing rule under the Sharpe ratio criterion. For each period, the table shows the “Reality Check” (RC) and the SPA p-values along with the corresponding nominal p-values. SPA-c refers to the consistent p-value of Hansen (2005)’s SPA; SPA-l is the lower bound of the consistent SPA p-value. The nominal p-values are obtained by applying the RC and the SPA testing procedures to the best market-timing rule only, without correcting for data-snooping biases.

Table 4: Best Market-Timing Rules in the Extended Set under the Mean Return Criterion and the Sharpe Ratio Criterion

Panel A Sample	Best Trading Rule	Mean Return Criterion		
		# Trades	Model Return	Benchmark Return
Full Sample (1981 - 2007)	Learning Rule, $r = 10, e = 1$	328	12.40%	9.22%
Subperiod I (1981 - 1994)	Learning Rule, $r = 5, e = 1$	370	24.24%	9.02%
Subperiod II (1995 - 2007)	Learning Rule, $r = 5, e = 5$	306	16.23%	9.35%

Panel B Sample	Best Trading Rule	Sharpe Ratio Criterion		
		# Trades	Model SR	Benchmark SR
Full Sample (1981 - 2007)	Learning Rule, $r = 10, e = 1$	328	0.60	0.237
Subperiod I (1981 - 1994)	Learning Rule, $r = 5, e = 1$	370	1.72	0.122
Subperiod II (1995 - 2007)	Learning Rule, $r = 5, e = 5$	306	1.11	0.350

This table reports the performance of the best market-timing rule under the mean return criterion (Panel A) and the Sharpe ratio criterion (Panel B) for each sample period. The Sharpe ratio is denoted by SR. The table also contains the mean return and the Sharpe ratio of the buy-and-hold strategy (benchmark model). The mean returns and the Sharpe ratios are annualized. Lower case letters r and e denote the length of the review span and of the evaluation period, respectively.

Table 5: Performance of the Best Market-Timing Rules in the Extended Set under the Mean Return Criterion

Sample	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.044	0.686	0.044	0.652	0.347
Subperiod I (1981 - 1994)	0.000	0.052	0.000	0.052	0.050
Subperiod II (1995 - 2007)	0.006	0.376	0.006	0.369	0.186

This table presents the performance of the best market-timing rule under the mean return criterion. For each period, the table shows the “Reality Check” (RC) and the SPA p-values along with the corresponding nominal p-values. SPA-c refers to the consistent p-value of Hansen (2005)’s SPA; SPA-l is the lower bound of the consistent SPA p-value. The nominal p-values are obtained by applying the RC and the SPA testing procedures to the best market-timing rule only, without correcting for data-snooping biases.

Table 6: Performance of the Best Market-Timing Rules in the Extended Set under the Sharpe Ratio Criterion

Sample	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.005	0.793	0.005	0.535	0.177
Subperiod I (1981 - 1994)	0.000	0.053	0.000	0.053	0.020
Subperiod II (1995 - 2007)	0.000	0.392	0.000	0.195	0.055

This table presents the performance of the best market-timing rule under the Sharpe ratio criterion. For each period, the table shows the “Reality Check” (RC) and the SPA p-values along with the corresponding nominal p-values. SPA-c refers to the consistent p-value of Hansen (2005)’s SPA; SPA-l is the lower bound of the consistent SPA p-value. The nominal p-values are obtained by applying the RC and the SPA testing procedures to the best market-timing rule only, without correcting for data-snooping biases.

Table 7: Performance of the Best Market-Timing Rules When Relaxing the Short-Selling Constraint

Sample	Panel A				
	Basic Set of Rules				
	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.033	0.698	0.033	0.675	0.461
Subperiod I (1981 - 1994)	0.037	0.176	0.037	0.163	0.114
Subperiod II (1995 - 2007)	0.089	0.779	0.089	0.784	0.629

Sample	Panel B				
	Extended Set of Rules				
	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.038	0.677	0.038	0.645	0.338
Subperiod I (1981 - 1994)	0.000	0.078	0.000	0.078	0.063
Subperiod II (1995 - 2007)	0.060	0.896	0.060	0.895	0.775

This table presents, for each sample period, the performance of the best market-timing rule under the mean return criterion when relaxing the short-selling constraint. Panel A includes the results when only simple trading rules are considered, whereas the results in Panel B are based on the extended universe of rules, including complex market-timing strategies. For each period, the table shows the “Reality Check” (RC) and the SPA p-values along with the corresponding nominal p-values. SPA-c refers to the consistent p-value of Hansen (2005)’s SPA; SPA-l is the lower bound of the consistent SPA p-value. The nominal p-values are obtained by applying the RC and the SPA testing procedures to the best market-timing rule only, without correcting for data-snooping biases.

Table 8: Performance of the Best Market-Timing Rules When Including Transaction Costs

Sample	Panel A				
	Basic Set of Rules				
	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.031	0.600	0.031	0.600	0.362
Subperiod I (1981 - 1994)	0.041	0.398	0.041	0.358	0.219
Subperiod II (1995 - 2007)	0.078	0.887	0.078	0.885	0.805

Sample	Panel B				
	Extended Set of Rules				
	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.072	0.808	0.072	0.768	0.456
Subperiod I (1981 - 1994)	0.000	0.056	0.000	0.056	0.050
Subperiod II (1995 - 2007)	0.020	0.510	0.020	0.500	0.250

This table presents, for each sample period, the performance of the best market-timing rule under the mean return criterion when including one-way transaction costs equal to 0.05%. Panel A includes the results when only simple trading rules are considered, whereas the results in Panel B are based on the extended universe of rules, including complex market-timing strategies. For each period, the table shows the “Reality Check” (RC) and the SPA p-values along with the corresponding nominal p-values. SPA-c refers to the consistent p-value of Hansen (2005)’s SPA; SPA-l is the lower bound of the consistent SPA p-value. The nominal p-values are obtained by applying the RC and the SPA testing procedures to the best market-timing rule only, without correcting for data-snooping biases.

Table 9: Performance of the Best Market-Timing Rules for Various Smoothing Parameters q .

Panel A					
Smoothing Parameter $q = 0.5$					
Sample	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.030	0.598	0.030	0.569	0.361
Subperiod I (1981 - 1994)	0.035	0.275	0.035	0.257	0.169
Subperiod II (1995 - 2007)	0.073	0.885	0.073	0.884	0.742
Panel B					
Smoothing Parameter $q = 0.1$					
Sample	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.026	0.558	0.026	0.539	0.309
Subperiod I (1981 - 1994)	0.058	0.254	0.058	0.235	0.131
Subperiod II (1995 - 2007)	0.076	0.879	0.076	0.872	0.717
Panel C					
Smoothing Parameter $q = 0.01$					
Sample	Reality Check		SPA		
	nom. p-value	RC p-value	nom. p-value	SPA-c	SPA-l
Full Sample (1981 - 2007)	0.047	0.518	0.047	0.494	0.319
Subperiod I (1981 - 1994)	0.044	0.168	0.044	0.154	0.072
Subperiod II (1995 - 2007)	0.122	0.857	0.122	0.828	0.609

This table presents, for each sample period, the performance of the best market-timing rule under the mean return criterion for alternative smoothing parameters q . The test results are based on the basic set of simple trading rules. For each parameter value, the table shows the “Reality Check” (RC) and the SPA p-values along with the corresponding nominal p-values. SPA-c refers to the consistent p-value of Hansen (2005)’s SPA; SPA-l is the lower bound of the consistent SPA p-value. The nominal p-values are obtained by applying the RC and the SPA testing procedures to the best market-timing rule only, without correcting for data-snooping biases.

Table 10: Summary of Applied Time Series

Name	Code	Description
S&P 500 Composite	S&PCOMP	Index Price
S&P 500 E/P	S&PCOMP(PE)	Earnings-to-Price Ratio - S&P 500
S&P 500 DY	S&PCOMP(DY)	Dividend Yield - S&P 500
S&P 500 B/M	N/A	Book-to-Market Ratio - S&P 500
Fed Funds Rate	FRFEDFD	US Federal Funds Rate (Effective) (% Per Annum)
Discount Rate	USFDTRG	US Federal Funds Target Rate (% Per Annum)
AAA Bond	Y70461	Moody's US AAA Corporate Bond (% Per Annum)
BAA Bond	Y70462	Moody's US BAA Corporate Bond (% Per Annum)
3-Mo. T-Bill	FRTBW3M	U.S.Treasury Bills, 2nd Market, 3-Mo.(% Per Annum)
6-Mo. T-Bill	FRTBW6M	U.S.Treasury Bills, 2nd Market, 6-Mo.(% Per Annum)
1-Yr. T-Bond	S02556	U.S.Treasury Const. Maturities 1-Yr. (% Per Annum)
2-Yr. T-Bond	S02557	U.S.Treasury Const. Maturities 2-Yr. (% Per Annum)
5-Yr. T-Bond	S02559	U.S.Treasury Const. Maturities 5-Yr. (% Per Annum)
10-Yr. T-Bond	S02561	U.S.Treasury Const. Maturities 10-Yr. (% Per Annum)
30-Yr. T-Bond	S02562	U.S.Treasury Const. Maturities 30-Yr. (% Per Annum)
CBOE Put/Call Ratio	PCUSSPXR Index	CBOE Put/Call Ratio
CBOE VIX	T59165	Implied Volatility Index for S&P 500 Index Options

Table 11: Parameter Values for Simple Rules

Indicator	Threshold Values
Financial Ratios	[0.02 : 0.01 : 0.12] for E/P [1 : 0.25 : 3] for DY [1 : 0.05 : 1.75] for B/M
Short-Term Interest Rates	[8 : 0.25 : 15]
Long-Term Interest Rates	[8 : 0.25 : 15]
Maturity Spread	[−3 : 0.25 : 1]
Expected Inflation	[0 : 0.25 : 1.5]
Credit Spread	[2 : 0.1 : 4]
Put/Call Ratio	[1 : 0.25 : 5]
Implied Volatility	[15 : 1 : 45]
Bond-Equity Yield Ratio	[1 : 0.05 : 1.8] for E/P and 3-Mo. T-Bill [1.5 : 0.05 : 2.4] for E/P and 10-Yr. T-Bond [2.5 : 0.25 : 5.5] for DY and 3-Mo. T-Bill [3 : 0.25 : 6] for DY and 10-Yr. T-Bond
Dividend Payout Ratio	[0.35 : 0.015 : 0.5]